Abdul Haseeb Shaikh

AI Engineer, RL Environments and LLM Evaluation

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SUMMARY

I am part of Turing's RL team in the Frontier AI Labs department, where for the last two years I have built the RL environments, verifiers and reward signals that frontier labs use to train and evaluate their models. I write the first environments on each project myself, then set the QC bar and review every task before it ships, more than 2,000 tasks and trajectories so far across evaluations of 10 models. Before that I worked in backend engineering from 2020, including leading software development at Cloud Mesh Solutions, mostly in Python, TypeScript, Docker and AWS. I compete in CTFs at a national level, which is why I design rewards and verifiers that are hard to game and quick to expose real model failures. I am looking for a remote role in RL environments, LLM evaluation or applied AI engineering.

SKILLS

RL environments: Agent environment design with tasks, data, tools and automated verifiers, reward design, reward hacking and answer leakage checks, solution and verifier parity, reproducible Docker environments

LLM evaluation: Benchmark and task design, failure mode analysis, rubric calibration, multi-turn trajectory review, SFT and RLHF data quality

Agentic AI: Tool-calling agents on the OpenAI and Anthropic APIs, structured outputs, prompt pipelines, sandboxed agent environments, RAG with hybrid retrieval (pgvector, Postgres full-text search, Reciprocal Rank Fusion), embeddings and retrieval evaluation

Eval tooling: Harbor tasks (task.toml, instruction.md, tests/test.sh, solution/solve.sh), BrowseComp-style web research tasks, SkillsBench, pytest, Docker, CI

Engineering: Python (Flask, Django), TypeScript and Node.js (Express), REST APIs, microservices, Docker, CI/CD, AWS EC2, Azure, GCP, Git

Security: Web pentesting, reverse engineering and CTF competition, which I use for adversarial and anti-leakage review of tasks

EXPERIENCE

Senior AI Engineer, RL Team (Frontier AI Labs) | Turing (remote, US based) — Nov 2024 to present

- Member of Turing's RL team in the Frontier AI Labs department. Build RL environments and evaluation infrastructure that frontier labs use to train and evaluate their models, across multi-turn software engineering, BrowseComp-style deep web research, SkillsBench, computer-use and tool-use environments, and an RL Gym benchmark for security tools.
- Helped build and run the RL Gym evaluation harness (FastAPI, Redis, PostgreSQL, Docker Compose) that sustains more than 2,000 agent runs a day across 10 to 15 web gyms for 480 users, and the platform that unified authoring, review and delivery, lifting weekly output from about 100 to 500 to 800 tasks.
- Built an RL Gym security benchmark scoring eight scanners (Semgrep, CodeQL, SonarQube and model-based reviewers) on sensitive-data and excessive-agency risk from the OWASP LLM Top 10, with 40 contrast tasks from real CVE advisories, SARIF normalization and 848 automated tests.
- Write the first Dockerized Harbor environments on each project myself, each with an agent-facing instruction, a reference solution and an automated verifier, and own the QC toolkits behind several projects. Reviewed more than 2,000 tasks and trajectories for verifier correctness, answer leakage and reward hacking, and built a closed-book BrowseComp authoring flow, review rubrics of up to 85 automated checks, and audits that verify tasker feedback against recorded trajectories and telemetry.
- Helped build five RL web environments as offline React apps with deterministic grading, evaluated 10 frontier models on reasoning, code understanding and tool use, and worked as Engineering Manager and Delivery Manager across a team of 60 engineers, grading SFT and RLHF data against lab rubrics.

Backend Engineer (contract, alongside Turing) | FuzionDev LLC, US (remote) — Dec 2025 to Mar 2026

- DrayMatch, a US container logistics platform. Built the Node.js and PostgreSQL backend and the scrapers that pull import and export manifests and draymen registry data, and specified the DrayTrends analytics layer (trade lanes, equipment mix, container-to-draymen index).
- Also shipped backends for Condo Whisper, FinanceIQ, ABCard, Soul Story and TimeCloud, mostly Node.js and TypeScript with Prisma or Sequelize, PostgreSQL, AWS S3 and SES, and Docker.

Backend Engineer (contract, alongside Turing) | AVDA.ai, AI voice agent platform for a US client — Sep 2025 to Jan 2026

- Designed and built the backend for a multi-tenant SaaS that runs AI phone agents for inbound and outbound calls. Real-time audio streams over WebSockets through Twilio Media Streams, speech goes through Google Cloud, AssemblyAI or ElevenLabs, and the conversation is driven by GPT-4o or Gemini with function calling for appointment booking, business hours and hand-off to a human.
- Node.js and TypeScript, MongoDB, Redis, AWS S3, Stripe subscriptions and minute credits, GoHighLevel CRM integration over OAuth, call recording, transcripts and an analytics dashboard. Interrupt detection keeps the agent responsive in under 300 ms when the caller talks over it. Solo build over roughly 170 commits.

Head of Software Development | Cloud Mesh Solutions (Pvt) Ltd, Islamabad — Nov 2022 to Dec 2024

- Led software development at a company of around 30 people, heading the engineering team across several SaaS and marketplace products for international clients. Owned system architecture, delivery, hiring and client-facing technical decisions.
- Products I led and coded on include Kaufes (a Swiss classifieds and auction marketplace with a Node.js backend, Next.js web app, Flutter mobile app, Wallee payments and Swiss QR bills), Park4Fly (airport parking management with a driver app), Büro 365 (task, appointment and project management for small businesses) and Offerten Helden (a Swiss quote and lead management platform for service businesses).
- Designed the backend systems in Node.js with TypeScript and Python with Django. Introduced clear service boundaries, containerization and CI/CD pipelines so that releases became routine instead of risky.
- Set the standards for code review, automated testing and observability, and mentored engineers into senior roles.

Lead Software Engineer (started as Software Engineer) | Ripple Intra Technologies, Karachi — Aug 2021 to May 2022

- Worked on apps for Concord Healthcare Group, including the Concord Pharma sales team app (backend and mobile), Toticell, EW Villa Medica and DMFR, owning backend and DevOps with Python (Flask) and TypeScript (Express), Docker-based microservices and AWS EC2. Towards the end I built the Brand Emirate backend (be-backend) in TypeScript with MongoDB and Redis, along with a shared microservices base layer.
- Promoted to lead in March 2022, where I set up the delivery pipelines and engineering workflow for the team.

Software Developer | Time Global — Jun 2020 to Dec 2020

- Built and maintained Python and Django backend systems, and improved the performance and maintainability of production services.

LEADERSHIP & COMMUNITY

President | CUI Tech Society, COMSATS University Islamabad — 2025 to 2026

- Led the CS department's umbrella student society and organized CUI Tech Fest 2025, a six-day national event with 1,200 participants across 12 modules.

Co-founder and President | Cyber Hacktivators Club (CHC), COMSATS University Islamabad — 2023 to 2025

- Founded the university's cybersecurity society, ran a CTF course and workshops, operated Docker-based CTFd infrastructure, and coordinated 30 teams for Black Hat MEA 2024, two of which reached the finals.

PROJECTS

advisory-rag, hybrid RAG over security advisories (personal project, 2026)

- Retrieval-augmented QA over 300 GitHub security advisories (github.com/haseebshaikh03/advisory-rag). Local BGE embeddings in PostgreSQL with pgvector (HNSW), vector and full-text results merged with Reciprocal Rank Fusion, and cited answers from Claude behind FastAPI. A 180-question retrieval eval (recall@5, MRR@10) showed that routing exact CVE ids to a direct lookup lifts recall@5 from 0.68 to 0.99.

ExploitOps, a multi-cloud attack simulator (final year project, 2025 to 2026)

- Offensive cloud security framework that simulates real attack paths across AWS, Azure and GCP in sandboxes provisioned with Terraform, covering bucket exploitation, serverless abuse, SSRF and metadata attacks, IAM privilege escalation and cross-cloud pivoting. Python and Flask with a CLI and a web dashboard, packaged with Docker. Two-person team.

EDUCATION

BS Cyber Security | COMSATS University Islamabad — 2022 to 2026

Completed while working full time in the engineering roles above. Final year project was ExploitOps (see Projects).

Diploma in Computer Science | Aptech Pakistan — 2019 to 2022

Three-year software development diploma, completed alongside school and my first developer roles.

Intermediate, Pre-Engineering | Government National College, Karachi — 2020 to 2022

ACHIEVEMENTS

- Organized CUI Tech Fest 2025, the largest event the CS department has run, with 1,200 participants across 12 competition, workshop and exhibition modules.
- Organized CodeCon-CUI'24, a three-day national event with 350 participants from 14+ universities, combining a 30-team CTF with gaming and coding competitions, run with Ignite and Trillium (COMSATS Annual Report 2024 to 2025, page 102).
- Black Hat MEA 2024 Finals, Riyadh. Placed 40th of 250 finalist teams with Team PsychoTherapist, after qualifying from a field of 9,000 teams.
- Ignite National Cyber Hackathon (Ministry of IT, 700 teams). 2nd place nationally in Red Teaming and 1st in the regional round, both rounds.
- DevConCTF'25 (NUST). 1st place nationally. Team PsychoTherapist is ranked 3rd in Pakistan on ctftime.org.